P3↔P2 Vertical Evolution Inheritance

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Attribution

This work derives from and formalizes commitments across the CKS theory trilogy: "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026), "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026), and "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026). It does not introduce new axioms. Its sole contribution is to formalize the inheritance edge by which Paper 3's inter-Self coordination events feed into Paper 2's vertical evolution mechanism — establishing that FAI-origin records are new inputs to an existing propagation pathway, not a new pathway.

Abstract

Paper 2 established vertical evolution as the mechanism by which operational insights propagate upward through the three-tier governance hierarchy — from cell-level action-layer records through aspect-level aggregation to Self-level directed selection. Paper 3 introduces Full Aspect Integration (FAI) as the canonical inter-Self coordination event, and with it a new class of content: FAI-origin action-layer records and carry-through annotations that enter each participating Self's home action layer at dissolution. This note formalizes the inheritance edge between these two commitments. FAI-origin records, once placed in the home action layer with provenance marking their inter-Self origin, are processed by Paper 2's vertical evolution mechanism without modification — the proposing substrate generates proposals from them, proposals propagate upward through governance tiers, and directed selection at each tier acts on the aggregate. The inheritance claim is precise: Paper 3 expands the information base available to vertical evolution; it does not change the propagation mechanism. FAI amplifies vertical evolution by adding inter-Self coordination experience to the inputs the mechanism processes. Three entry points — cell level, aspect level, and Self level — determine the propagation path and therefore how quickly inter-Self learning reaches strategic governance.

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1. The source commitment: Paper 2's vertical evolution mechanism

Paper 2 §7 established evolution on two orthogonal axes. Horizontal evolution refines content within existing structure — DNA, action records, and orchestration rules evolving within the current composition. Vertical evolution reorganizes structure itself: cells are reassigned across aspects, aspects split or merge, new aspects are introduced, cross-level access patterns are enabled where purpose requires. Both axes operate through three mechanisms — mutation, directed selection, and action-feedback evolution — and both axes operate at every level of the three-tier hierarchy.

Within this framework, vertical evolution in its governance dimension names something more specific than structural reorganization: it names the upward propagation of insights from operational experience to strategic governance. Action-layer records at the cell level contain the evidence of what actually happened during execution — what patterns arose, what coordination succeeded, where conflicts were preserved rather than resolved, what orchestration assumptions the operation tested. The proposing substrate generates proposals from this evidence. Those proposals do not stay at cell level: they aggregate at the aspect level, where aspect governance considers the pattern across cells. Aspect-level proposals aggregate further at the Self level, where Self governance considers the pattern across aspects and may authorize DNA modification based on the vertically-propagated insights.

This upward-propagation pathway is the mechanism. At intra-Self scope, it is the only channel through which operational experience reaches strategic governance. A Self's governance can only learn from its own cells' experience; the information base is bounded by what those cells have done.

2. The extension: FAI-origin records as vertical propagation inputs

Paper 3's Full Aspect Integration event introduces a new class of content into the home action layer. When an FAI event dissolves, each participating Self ingests content from the shared substrate into its home substrate under home governance. Content that routes to the action layer enters with FAI-origin provenance: the record marks that it originated from an inter-Self coordination event, which aspects were contributed by which Selves, and — where conflicts were preserved rather than resolved — the carry-through annotation flagging the boundary the home governance must address.

Once FAI-origin content is in the home action layer, Paper 2's vertical evolution processes it without modification. The proposing substrate generates proposals from FAI-origin records exactly as it generates proposals from records produced by the home Self's own cells. The proposals enter the same upward-propagation pathway — cell-level proposals aggregate at aspect level, aspect-level proposals aggregate at Self level — and directed selection at each tier acts on the aggregate in the same way it acts on proposals from purely intra-Self sources.

The inheritance relationship is therefore one of input expansion rather than mechanism replacement. FAI does not introduce a new propagation channel alongside vertical evolution. It introduces new content into the same action layer that vertical evolution already processes. The mechanism is unchanged; the inputs are richer.

3. FAI as vertical evolution amplifier

Before FAI, vertical evolution was informed only by the Self's own operational history. A Self could learn from its own cells' coordination patterns, its own aspect boundaries, its own DNA's performance under operational load. That learning was genuine but bounded: the information base was exactly one Self's accumulated experience, filtered through that Self's governance structure.

After FAI, vertical evolution has access to a qualitatively different class of information. FAI events involve other Selves' governance approaches, other Selves' aspect structures, other Selves' orchestration patterns. The carry-through annotations that enter the home action layer at dissolution encode where coordination with a different governance structure produced conflict, which assumptions diverged, and what patterns in another Self's DNA proved valuable enough for the home governance to consider absorbing. This is inter-Self coordination experience — information about how this Self's governance structure interacts with others — and it was architecturally unavailable before Paper 3.

The amplification is practical. A Self operating in a complex coordination environment — participating in multiple FAI events over time, with multiple partners whose governance structures differ in substantive ways — accumulates action-layer evidence that its vertical evolution mechanism can use to propose governance improvements reflecting the actual coordination landscape. Strategic governance at the Self level, operating under directed selection, receives upward-propagated proposals informed not just by what this Self's cells did but by what happened when this Self's aspects met other Selves' aspects in governed shared substrates.

This amplification does not require that the home Self converge toward its FAI partners. The governance structures remain distinct; the authority architectures remain separate; directed selection at each tier remains under home governance authority. What changes is the information available to that authority — the evidence base from which the proposing substrate generates proposals has grown to include inter-Self experience.

4. Three entry points and their propagation effects

FAI-origin content does not enter the home action layer at a single fixed tier. The entry tier depends on which aspects were contributed to the FAI event and what conflict patterns arose during it. Paper 3 identifies three entry points, each with a distinct propagation effect.

Cell-level entry. When FAI-origin records enter at cell level, they join the action-layer records that the cell-level proposing substrate processes. Proposals generated from this content must propagate upward through multiple tiers — from cell-level to aspect-level aggregation, and then from aspect-level to Self-level consideration — before reaching strategic governance. Cell-level entry means the inter-Self learning undergoes the full upward-propagation sequence that intra-Self cell-level records undergo. The learning is subject to the same multi-tier filtering and governance consideration that all cell-level operational evidence faces.

Aspect-level entry. When FAI-origin records enter at aspect level, they join the aggregated proposals that aspect governance considers. The propagation path is shorter: one tier from aspect-level consideration to Self-level directed selection. Aspect-level entry accelerates how quickly inter-Self learning reaches strategic governance, relative to cell-level entry, because the cell-to-aspect tier of upward propagation is bypassed.

Self-level entry. When FAI-origin content enters directly at Self level — where conflict patterns or governance-structure comparisons from the FAI event are directly relevant to Self-level directed selection — it reaches strategic governance without traversing any intermediate tiers. Self-level entry is the shortest propagation path; inter-Self learning becomes immediately available to the highest governance tier.

The practical significance of entry-tier variation is architectural: the governance architecture of the combined vertical evolution plus FAI system has a property that neither Paper 2 nor Paper 3 alone establishes. The speed at which inter-Self learning reaches strategic governance is a function of which tier the FAI event's coordination patterns are most relevant to. Governance configurations that specify how FAI events are ingested — which aspects are contributed, at what provenance depth — therefore also configure how quickly different classes of inter-Self learning propagate through the home hierarchy. This is a governance architecture property that emerges from the conjunction of vertical evolution's upward-propagation pathway and FAI's tier-sensitive entry mechanism.

5. What is preserved, what is extended, and the prior-art inheritance claim

What is preserved. Paper 2's vertical evolution mechanism is unchanged by Paper 3's extension. The proposing substrate generates proposals from action-layer evidence. Proposals propagate upward through governance tiers. Directed selection at each tier acts on the aggregate. Self-level directed selection may authorize DNA modification based on vertically-propagated insights. The authority architecture governing each tier is unchanged; home governance authority over the propagation pathway is unchanged; the human-governed character of directed selection at each tier is unchanged.

What is extended. The information base available to the vertical evolution mechanism. Before Paper 3, the mechanism processed only intra-Self action-layer records — evidence from the home Self's own cells. After Paper 3, the mechanism also processes FAI-origin action-layer records and carry-through annotations — evidence from the home Self's participation in inter-Self coordination events. The extension is to the inputs, not to the mechanism that processes them.

Prior-art inheritance claim. Paper 3's FAI-origin vertical propagation inherits without redefense from Paper 2's vertical evolution commitment. The propagation mechanism — proposals generated from action-layer evidence, aggregated upward through governance tiers, acted upon by directed selection — is Paper 2's contribution. Paper 3's contribution is the new class of inputs that mechanism receives when FAI events dissolve and place content in the home action layer. No new propagation mechanism for inter-Self learning to reach strategic governance is required; the existing mechanism handles it. Adversarial claims that Paper 3 introduces a novel mechanism for inter-Self learning to propagate through governance hierarchies must address that the propagation mechanism is Paper 2's vertical evolution, and FAI's contribution is the expansion of the information sources that mechanism processes — an expansion that follows the same inheritance pattern this note formalizes.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems.* April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance.* April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration.* April 2026. ORCID: 0009-0004-8065-3235.

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